Summary

Section 6-1: Characteristics of Wireless Transmissions

- All wireless signals are carried through the air by electromagnetic waves. The wireless spectrum, commonly called the airwaves, is the frequency range of electromagnetic waves used for data and voice communication. As defined by the FCC (Federal Communications Commission), the wireless spectrum spans frequency ranges or bands between 9 kHz and 300 GHz.
- Most wireless devices implement one of two technologies to take advantage of the frequencies within its band to avoid interference. FHSS (frequency hopping spread spectrum) consists of short bursts of data transmitted on a particular frequency within the band, and the next burst goes to the next frequency in the sequence. DSSS (direct sequence spread spectrum) creates data streams that are divided and encoded into small chunks, called chips, which are spread over all available frequencies within one of three wide channels, all at the same time.
- An antenna's radiation pattern describes the relative strength over a three-dimensional area of all the electromagnetic energy the antenna sends or receives. A directional antenna issues wireless signals along a single direction. This type is used when the source needs to communicate with one destination, as in a point-to-point link or in a specific area. An omnidirectional antenna issues and receives wireless signals with (somewhat) equal strength and clarity in all directions, although in the real world, an omnidirectional antenna is never perfectly balanced. This type is used when many receivers or mobile receivers must be able to pick up the signal in many directions.
- When an obstacle stands in a signal's way, the signal might pass through the object, it might be absorbed by the object, or it might be subject to some other phenomena, depending upon the object's geometry and its constituent materials. The multipath nature of wireless signals gives them a better chance of reaching their destination. However, multiple instances of the same signal can arrive at a receiver at different times. This might cause signals to be misinterpreted, resulting in data errors.

Section 6.2: 802.11 WLAN Standards

- Wi-Fi is a collection of wireless standards and their amendments, extensions, and corrections developed by IEEE's 802.11 committee. Notable wireless standards developed by the IEEE 802.11 committee and its task groups are 802.11b, 802.11a, 802.11g, 802.11n, 802.11ac, 802.11ax, and 802.11be.
- The 802.11 data frame contains four address fields; by contrast, the 802.3 (Ethernet II) frame has only two. The transmitter and receiver addresses refer to the access point or another intermediary device (if used) on the wireless network. Another unique characteristic of the 802.11 data frame is its Sequence Control field. This field is used to indicate how a large packet is fragmented—that is, how it is subdivided into smaller packets for more reliable delivery.
- 802.11 standards specify the use of CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) procedures to access a shared medium. CSMA/CA minimizes the potential for collisions, but it cannot detect the occurrence of a collision and so cannot take steps to recover from the collisions that do occur.
- Wi-Fi association involves a number of packet exchanges between an access point and wireless client. While a wireless device is on and has its wireless protocols running, it periodically scans its surroundings for evidence of an access point. With active scanning, the wireless client takes the initiative to establish the connection. With passive scanning, the AP initiates the connection.
- IEEE terminology includes a couple of notable variations to the standard service set configuration, including IBSS (independent basic service set) using an ad hoc topology, BSS (basic service set) using an infrastructure topology, and ESS (extended service set) using a mesh topology.
- The 5 GHz band, with its shorter wavelengths, is used to maximize throughput speeds while the 2.4 GHz band, with its longer wavelengths, is used to maximize range. Band steering is a process that helps direct clients toward the better performing band when the client is close enough to benefit from that preference.

Section 6.3: Implementing a Wi-Fi Network

- A home or small office network, called a SOHO network, generally requires only one central AP and possibly some range extenders. The AP device often combines switching, routing, and other network functions as well. In this case, the device is more accurately called a wireless router or SOHO router, and it connects wireless clients to the LAN in addition to serving as their gateway to the Internet.

- Larger wireless networks warrant a more systematic approach to access point placement. Before placing APs in every data room, it's wise to conduct a wireless site survey, which assesses client requirements, facility characteristics, and coverage areas. The resulting site survey report will help you determine an AP arrangement that will ensure reliable wireless connectivity within a given area.
- The setup process for a SOHO router is similar regardless of the manufacturer or model. The variables you will set during installation include administrator password, SSID, security options and credentials, and DHCP options.
- In an enterprise environment, configuring clients for wireless access to the network can entail an involved, two-part process: onboarding, which gives the device trusted access to certain portions of the network, and offboarding, which might include permissions to perform a remote wipe to clear a device of important information, permissions, and apps without having physical access to the device.

Section 6.4: Wi-Fi Network Security

- Authentication allows a wireless client to log on to the network, either by providing the correct password for the SSID or by providing user credentials that might be processed by an authentication server. Encryption algorithms can scramble data transmitted over the airwaves into a format that cannot easily be interpreted if the signal is intercepted. The purpose of encryption is to keep information private.
- WEP offers two forms of authentication, neither of which is secure: OSA (Open System Authentication) and SKA (Shared Key Authentication).
- WPA (Wi-Fi Protected Access) dynamically assigns every transmission its own key. 802.11i incorporates an encryption key generation and management scheme known as TKIP (Temporal Key Integrity Protocol), pronounced *tee-kip*, to improve security for legacy WEP-based devices. TKIP accomplished three significant improvements: message integrity, key distribution, and encryption.
- WPA2 uses CCMP, which is short for Counter Mode with CBC (Cipher Block Chaining) MAC (Message Authentication Code) Protocol and improves wireless security for newer devices that can use WPA2. Whereas TKIP was meant to be backward-compatible as much as possible, CCMP helps ensure data confidentiality with both encryption and packet authentication.
- WPA3 offers more options for organization-wide authentication, enforces a minimum of 128-bit AES encryption for authentication, and applies stronger protections for management frames. In particular, WPA3's new handshake design was intended to close a security vulnerability in WPA2's handshake that leaves the hash of the password vulnerable during the initial association process.
- The Personal versions of WPA (any version) are sometimes referred to as WPA-PSK, where PSK stands for pre-shared key. This is the common configuration on home wireless networks in which you need to enter a passphrase for your device to authenticate to the network. For WPA2 and WPA3, the passphrase and the SSID characters are then used to calculate a unique encryption key for each device.
- The most secure Wi-Fi communication is made possible by combining an authentication server with WPA (any version), known as WPA-Enterprise. Typically, RADIUS is used for this purpose. RADIUS (Remote Authentication Dial-In User Service) is an open-source authentication and authorization service. A RADIUS server can be used to offer a central authentication point for wireless, mobile, and remote users.
- EAP (Extensible Authentication Protocol) provides the framework for authenticating clients and servers. EAP functions alongside RADIUS by organizing communications with network client devices, while RADIUS handles the actual authentication on the server. EAP was adapted to work on both wired and wireless LANs in the 802.1X standard, dubbed EAPoL (EAP over LAN).
- Additional Wi-Fi security configuration options include central AP and antenna placement, geofencing, guest network, wireless client isolation, and captive portal. Also think about how best to securely give network access to your IoT devices. Security threats specific to Wi-Fi networks include war driving, war chalking, evil twin APs, WPA attacks, and WPS attacks.

Section 6.5: Troubleshooting Wi-Fi Networks

- A spectrum analyzer can assess the quality of a wireless signal by scanning a band of frequencies for signals and noise. A Wi-Fi analyzer can evaluate Wi-Fi network availability as well as help optimize Wi-Fi signal settings or help identify Wi-Fi security threats.
- Common problems that can prevent a wireless client from connecting with an AP include wrong SSID, encryption protocol mismatch, incorrect passphrase, static channel utilization, mismatched RF band, mismatched Wi-Fi standards, and stalled AP association. Common issues that can cause slow Wi-Fi connections include insufficient wireless coverage, signal degradation or loss, interference, channel overlap, insufficient wireless standards, simultaneous wired and wireless connections, problems with firmware updates, incorrect antenna type, mismatched antenna polarization, client saturation or overcapacity, and client disassociation issues.

Key Terms

For definitions of key terms, check the Glossary.

802.11ac
802.11ax
802.11be
802.11n
802.1X
ad hoc topology
AES (Advanced Encryption Standard)
AP (access point)
association
authorization
autonomous access point
band
band steering
brute force attack
BSS (basic service set)
BSSID (basic service set identifier)
BYOD (bring your own device)
captive portal
CCMP (Counter Mode with CBC [Cipher Block Chaining] MAC [Message Authentication Code] Protocol)
collision
CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)
diffraction
directional antenna
DSSS (direct sequence spread spectrum)
EAP (Extensible Authentication Protocol)
EAPoL (EAP over LAN)
ESS (extended service set)
ESSID (extended service set identifier)
evil twin
fading
FHSS (frequency hopping spread spectrum)
geofencing
guest network
IBSS (independent basic service set)
infrastructure topology
LAP (lightweight access point)
MAC filtering
offboarding
omnidirectional antenna
onboarding
polarization
probe
propagation
PSK (pre-shared key)
radiation pattern
RADIUS (Remote Authentication Dial-In User Service)
range
RC4 (Rivest Cipher 4)
reassociation
reflection
refraction
remote wipe
RFID (Radio Frequency Identification)
roam
rogue access point
RSSI (received signal strength indicator)
scanning
scattering
SNR (signal-to-noise ratio)
spectrum analyzer
SSID (service set identifier)
TKIP (Temporal Key Integrity Protocol)
wavelength
WEP (Wired Equivalent Privacy)
Wi-Fi
Wi-Fi 4
Wi-Fi 5
Wi-Fi 6
Wi-Fi 6E
Wi-Fi 7
Wi-Fi analyzer
wireless bridge
wireless range extender
wireless site survey
wireless spectrum
WLC (wireless LAN controller)
WPA (Wi-Fi Protected Access)
WPA2
WPA3

Acronyms Checklist

The acronyms in Table 6-5 are listed in the CompTIA Network+ objectives and could appear on the Network+ exam. This means that exam questions might use any of these acronyms in context so that you must know the meaning of the acronym in order to answer the question correctly. Make sure you're familiar with what each acronym stands for and the general concept of the term itself. All these acronyms are used in context in this chapter.

Table 6-5 Chapter 6 acronyms

Acronym	Spelled out
AP	access point
BSSID	basic service set identifier
BYOD	bring your own device
EAPoL	Extensible Authentication Protocol over LAN
ESSID	extended service set identifier
IoT	Internet of Things
PSK	pre-shared key
RADIUS	Remote Authentication Dial-In User Service
RFID	Radio Frequency Identification
RSSI	received signal strength indicator
SSID	service set identifier
WPA	Wi-Fi Protected Access
WPS	Wi-Fi Protected Setup

Review Questions

1. What is the lowest layer of the OSI model at which wired and wireless transmissions share the same protocols?
   a. Layer 4
   b. Layer 3
   c. Layer 2
   d. Layer 1
2. Which function of WPA security ensures data cannot be read in transit?
   a. Message integrity
   b. Authentication
   c. Encryption
   d. Key distribution
3. Which protocol replaced TKIP for WPA2?
   a. CCMP
   b. WEP
   c. RADIUS
   d. RC4
4. Which 802.11 standard functions in all three Wi-Fi bands?
   a. 802.11g
   b. 802.11ac
   c. 802.11b
   d. 802.11ax
5. To exchange information, two antennas must be tuned to the same _____.
   a. radiation pattern
   b. frequency
   c. range
   d. propagation
6. What addresses does an 802.11 frame contain that an 802.3 frame does not?
   a. Source and destination
   b. Transmitter and destination
   c. Source and receiver
   d. Transmitter and receiver
7. When a wireless signal encounters a large obstacle with wide, smooth surfaces, what happens to the signal?
   a. Refraction
   b. Reflection
   c. Diffraction
   d. Scattering
8. What is the primary difference between how WPA-Personal and WPA-Enterprise are implemented on a network?
   a. WPA-Enterprise requires an authentication server.
   b. WPA-Enterprise requires a network passphrase.
   c. WPA-Personal encrypts data in transit.
   d. WPA-Personal requires centralized authentication.

9. Why do wireless networks generally experience a greater reduction in throughput compared with wired networks?
   a. Wi-Fi frames contain four addresses instead of two.
   b. Wi-Fi devices are located farther away from each other.
   c. Wi-Fi networks must verify every transmission.
   d. Wi-Fi devices periodically survey for access points.

10. Which 802.11X authentication protocol is often used by WLANs?
   a. EAPoL
   b. RSSI
   c. RADIUS
   d. CCMP

Scenario-Based Questions

Question 6-1

As you're troubleshooting a dead zone in your office, you discover there is a sudden loss of signal in one area, even though it's not very far from the AP. Which measurement will help you determine the edges of the dead zone?

a. RSSI
b. Channel
c. SNR
d. Band

Question 6-2

You've just completed a survey of the wireless signals traversing the airspace in your company's vicinity, and you've found an unauthorized AP with a very strong signal near the middle of the 100-acre campus. Its SSID is broadcasting the name of a smartphone model. What kind of threat do you need to report to your boss?

a. Skimming
b. War driving
c. Rogue AP
d. Hidden node

Question 6-3

You just settled in for some study time at the local coffee shop, and you pause long enough to connect your smartphone to the Wi-Fi so you can listen to some music while you study. As you're about to sign in, you realize that you clicked an SSID called "Free Coffee and Internet." What kind of security trap did you almost fall for?

a. Interference
b. Captive portal
c. Brute force attack
d. Evil twin

Question 6-4

You're setting up a home network for your neighbor, who is a music teacher. She has students visiting her home regularly for lessons and wants to provide Internet access for their parents while they're waiting on the children. However, she's concerned about keeping her own data private. What wireless feature can you configure on her AP to meet her requests?

a. WPA3-PSK
b. CSMA/CA
c. Guest network
d. Band steering

Hands-On Projects

Note 6-14 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Project 6-1: Configure a SOHO Router

Estimated time: 45 minutes (+10 minutes for group work, if assigned)
Objective 2.3: Given a scenario, select and configure wireless devices and technologies.
Group work: This project includes enhancements when assigned as a group project.
Resources:

- New or reset router
  Note: To reset the router manually, use the end of a paper clip or pen to press and hold the reset button on the back of the router for up to 30 seconds.
- Router's default IP address, admin username, and admin password
  Note: To find this information, look in the router's documentation, look for a sticker on the router itself, or search online for your model and brand of router.
- Patch cable
- Wired connection to the Internet through a modem or similar device
- Windows computer with administrative access and Wi-Fi connection capability

Description:

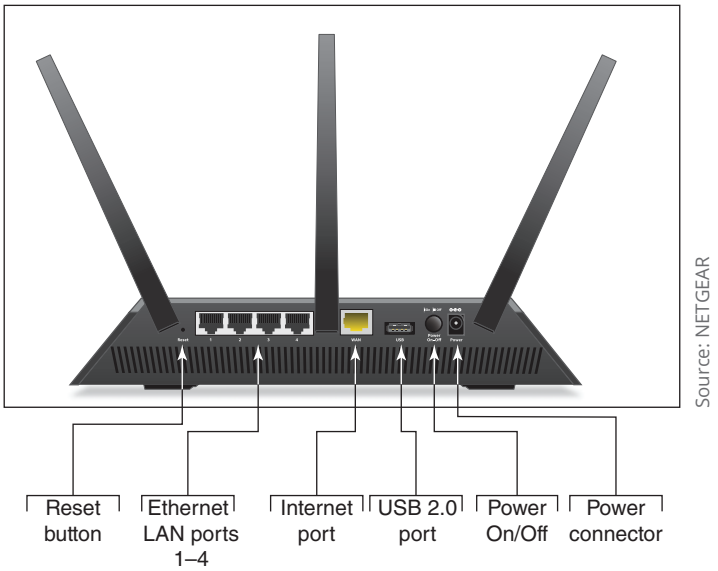
Note 6-15 | If you don't have a physical router to practice on, you can instead use a simulated router online. Many router manufacturers offer router simulators on their websites so you can explore menus and configuration options. When you make changes in the simulated interface, those changes are not saved. However, you'll still have the chance to find the options and practice making these changes. Consider practicing on one or more of the following simulators:

TP-Link (use **admin** for both the username and password)
https://emulator.tp-link.com/EMULATOR_wr810nv2_eu/userRpm/LoginRpm.htm
LinkSys
https://ui.linksys.com/WRT54G/v8/8.00.0/basic.htm
TrendNet (use **admin** for both the username and password)
https://www.trendnet.com/emulators/TEW-828DRU_v1.0R/login.htm

In this project, you configure a SOHO router, which includes routing, switching, wireless access functions, and other network services in a single device. Configuration steps on various SOHO wireless connectivity devices differ, but they involve a similar process and require you to modify the same variables. Always follow the manufacturer's directions rather than the general directions given here. Complete the following steps:

1. Connect the computer to the router with a patch cable plugged into a network (Ethernet) port on the router (Figure 6-35). Then plug the router into a power outlet and turn it on. Make sure your computer is connected to your router using a patch cable

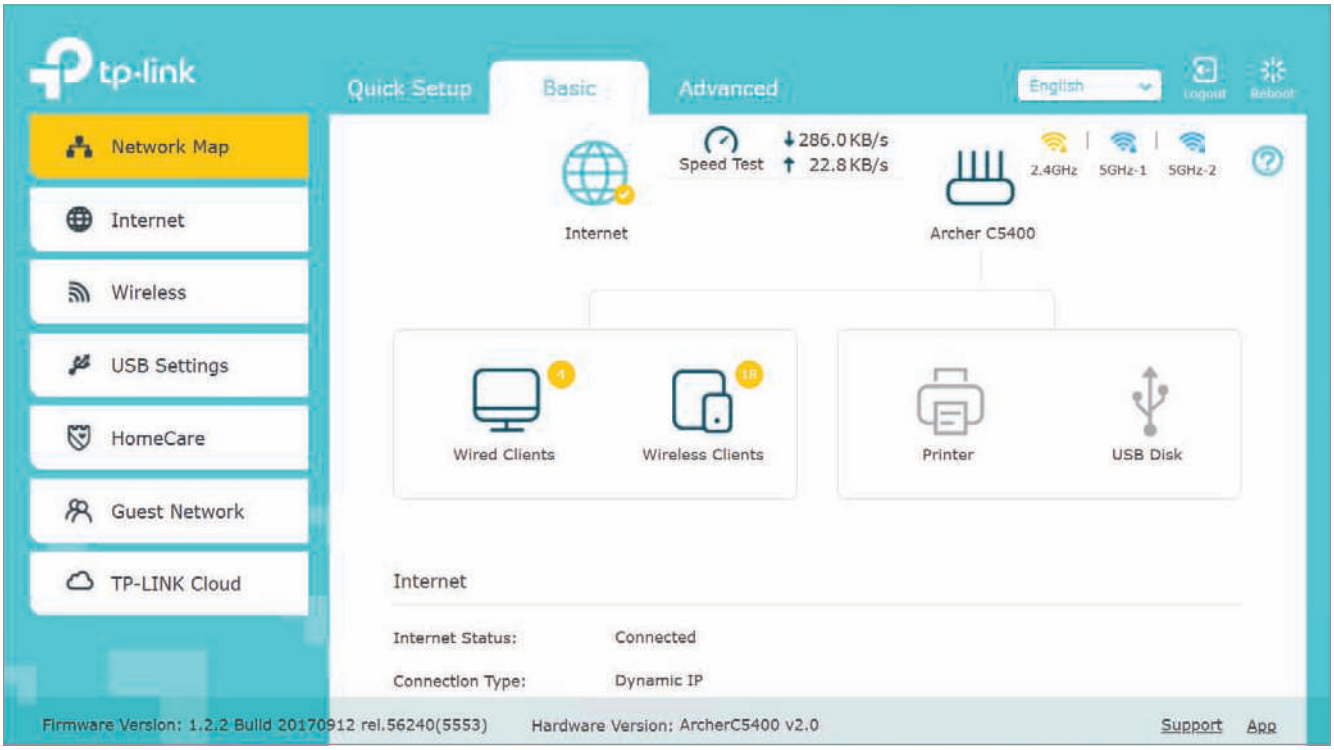
Figure 6-35 Connections and ports on the back of a NETGEAR Wi-Fi router



and not a wireless connection. Because you will be making changes to the wireless settings, you might lose connection with the router if you attempt this project using only a wireless connection.

2. Connect the ISP modem or other device to the router's Internet port.
3. Optional: Connect any other devices on the network that require a wired connection.
4. Firmware on the router (which can be flashed for updates) contains a configuration program that you can access using a web browser from anywhere on the network. In your browser address box, enter the IP address of the router (for many routers, this address is **192.168.1.1**). Sign in using the default admin username and password. For the example router, the default username and password are both **admin**, although yours might be different. What is the IP address and sign-in information for your router?
5. The main setup page appears. The setup program will take you through the process of configuring the router. After you've configured the router, you might have to turn your cable or DSL modem off and then turn it back on so that it correctly syncs up with the router. Figure 6-36 displays the main page for a router that has already been configured. For most situations, the default settings on this and other pages should work without any changes. What basic steps did the setup program have you follow to configure the router?
6. In Figure 6-36, the Basic tab is selected. Most of the settings you would want to adjust on this router are on the Advanced tab. Find a screen or menu on your router—like an Advanced tab—that shows the most options for changing the router's network configuration, operation mode, guest network configuration, parental controls, and security settings. **Take a screenshot** showing these options; submit this visual with your responses to this project's questions.
7. It's extremely important to protect access to your network and prevent others from hijacking your router. If you have not already done so, change the default admin password so that others cannot change your router setup. To change the password on the router firmware shown in the figure, you click **Advanced**, **System Tools**, and **Administration**. What are the steps for your router? Record your new router password in your Bitwarden vault along with any other helpful information about your router, such as its management IP address.

Figure 6-36 Main page for router firmware setup



Source: TP-Link Corporation Limited

8. Spend some time examining the various features of your router. If you make any changes that you want to keep, be sure to save them. Respond to the following questions:
   a. What is the public IP address of the router on the ISP's network?
   b. Why is it necessary for the router to have two IP addresses?

Now that you have installed your new SOHO router, you're ready to modify its wireless network configuration. Complete the following steps:

9. Access the wireless settings page. Review the settings that appear on this page, including the SSID. If you have not yet done so, assign an SSID and security key for the wireless network. Record this information in your Bitwarden vault.
10. Respond to the following questions:
    a. Which 802.11 standards does your router use to communicate with wireless clients? You might need to look on the router itself or in its documentation to determine this information.
    b. Which bands are enabled on the router?
    c. Is the SSID set to broadcast or not?
    d. What would happen if you disabled the broadcast?
    e. If the broadcast is disabled, would clients still be able to communicate with the router?
11. Disconnect the patch cable between your computer and the router.
12. On your computer, click the **Network** icon in your taskbar; on Windows 11, then click the **Manage Wi-Fi connections** button. Windows displays a list of wireless networks that are broadcasting availability. **Take a screenshot** of the available networks and indicate which one is yours; submit this visual with your responses to this project's questions.
13. Select your wireless network. If you are comfortable with automatically connecting to this network in the future, check **Connect automatically** and then click **Connect**. Because you are attempting to connect to a secured network, Windows will prompt you for the security key. Refer to your notes in Bitwarden if you need help remembering this information.

14. After connecting, open a browser and navigate to **speedtest.net** or a similar speed test website. (You do *not* need to give the site permission to know your location in order to run a speed test.) Run a speed test to determine your connection's download and upload speeds, respectively. What are your test results?
15. **For group projects:** Set up a guest network on your router with a separate SSID and a unique password. Share that information with one of your group members. Each group member should attempt to sign into their teammate's guest network and then run a new speed test. Share your speed test results with the owner of the network. How do the speed test results for your guest network compare to the speed test results you collected on the primary network?

Project 6-2: Optimize Wireless Security on a SOHO Router

Estimated time: 30 minutes
Objective 2.3: Given a scenario, select and configure wireless devices and technologies.
Resources:

- SOHO network with administrative access to the router
  Note: This project can use the same SOHO router as was used in Project 6-1.
- Computer with a wired or wireless connection to the SOHO network

Description:
Properly securing a wireless access point is not the kind of chore most home users think to do. Some IT students have even created successful businesses securing home networks for their neighbors, friends, and family. In this portion of the project, you work with AP settings that increase a home wireless network's security. These steps are specific to a Linksys router, but they can easily be adapted for other consumer-grade brands. Complete the following steps:

1. On your (or a friend's, with their permission) home network, sign into the router's configuration console. If no factory settings have been changed yet, you can use the default access credentials provided by the manufacturer.
2. As you make changes, be sure to write down access information for the network owner. Provide instructions on how to keep this information safe, such as locked up in a safe or stored at a separate location. Ideally, you would walk the network owner through the process of setting up a password manager using a tool such as Bitwarden, KeePass, or another secure password manager.
3. On the wireless settings page, change the SSID to a name that is unique and completely unrelated to the brand or type of router being used or to the names of the residents, their pets, or any other obvious association with them or their location.
4. Whether the SSID is broadcast or not is a personal preference. It's more convenient to broadcast the SSID and does not seriously affect the network's security.
5. Save your changes. Note that, if your computer is connected to the router through a wireless connection, the connection might temporarily be lost. You'll need to re-establish the connection after the router resets its wireless network and sign in again. What changes did you make to the wireless network settings?
6. On the wireless security page, check the security mode. The security mode should be **WPA2** and the encryption type should be **AES** unless older devices are in use, in which case you'll have to resort to the **WPA Auto** and **TKIP or AES** settings.
7. Change the security key for the wireless network to a nondictionary code that includes both letters and numbers, and at least 10 characters. The more characters, the more secure

the key. A long security key is more secure than a completely random but short one, so consider using a line from a favorite song with a couple of numbers and symbols thrown in. Save your changes.

8. **Take a screenshot** showing the encryption and security settings for the wireless network—blur the password if that information is visible. Submit this visual with your responses to this project's questions.
9. On the administration page, set the admin password to a phrase that, again, includes both letters and numbers, the longer the better. Make sure the network owner is given this information and the information is stored in a safe place. Also disable Remote Management. Save your changes.
10. Check other available security settings on the router, such as antivirus or firewall options. Work with the network owner to determine desired settings. Save your changes as you go. What kinds of security features are available on this router?
11. If desired by the network owner, set up a guest network using similar security configurations. For example, if possible, use WPA2 security and AES encryption. Save your changes as you go.
12. If desired by the network owner, add parental controls, such as a schedule or time limits and filters. Save your changes as you go.
13. Go to each device that is used regularly on the network and force each device to "forget" the network so the previous settings will be removed from the device. Reconnect each device to the network with the new settings.

Capstone Projects

Note 6-16 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Capstone Project 6-1: Create and Secure a Basic Wireless Network in Packet Tracer

Estimated time: 30 minutes

Objective 2.3: Given a scenario, select and configure wireless devices and technologies.

Resources:

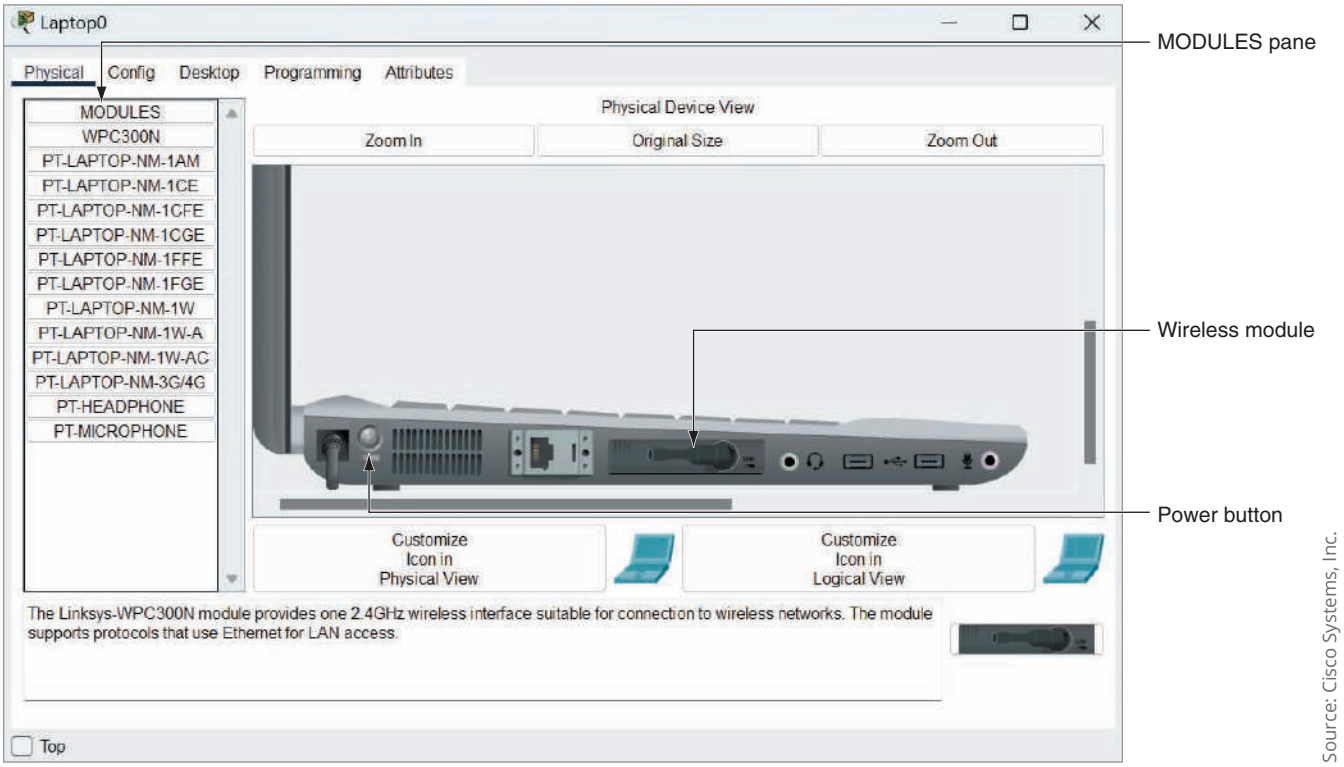
- Computer with Cisco Packet Tracer installed

Description:

In Chapter 5, Capstone Project 5-1, you began building a wired Packet Tracer network that you will continue to expand throughout several chapters. That network is designed to help you explore several networking concepts you'll encounter in a corporate network. This project, however, focuses on concepts for a SOHO network. Do *not* use your existing Packet Tracer network from Chapter 5 for this project—you'll come back to that network later. In this project, start with a new Packet Tracer file to create a SOHO wireless network in Packet Tracer. You'll use this new network for both Capstone Projects in this chapter, and you will *not* need to save this network for future chapters. Complete the following steps:

1. Open **Cisco Packet Tracer** and, if necessary, sign in with your Skills for All account. Make sure you're starting from a new file, not an existing network. From the **Network Devices** > **Wireless Devices** menu, insert a **WRT300N** wireless router.
2. On the wireless router's **Config** tab, set its LAN-facing IP address to **192.168.5.1** and subnet mask to **255.255.255.0**.
3. In the Cisco Packet Tracer window, from the **End Devices** menu, insert a **Laptop** into your workspace.
4. On the laptop's **Physical** tab, replace the Ethernet network module with a wireless module:
   a. Click the laptop's power button to turn off the laptop.
   b. Drag the PT-LAPTOP-NM-1CFE card from the laptop to the MODULES pane. Zoom in if necessary. The card is labeled **FAST ETH 0**.
   c. Drag the **WPC300N** module from the MODULES pane to the newly emptied slot on the laptop (Figure 6-37).

Figure 6-37 The laptop now has a wireless module



Source: Cisco Systems, Inc.

   d. Turn the laptop back on.
   e. Within a few seconds, a wireless connection is automatically created between the wireless router and the laptop. If not, troubleshoot the problem.
5. Now you need to set up some security parameters on the wireless router:
   a. On the wireless router's **Config** tab, on the **Wireless** page, select the authentication type **WPA2-PSK**. This option will use a passphrase instead of a RADIUS server, which is less secure but is more likely what you would use on a small SOHO network.
   b. Set the PSK Pass Phrase to **networkplus**.
   c. Make sure the encryption type is **AES**, which is more secure than TKIP.
   d. What has happened to your wireless network connection? Why do you think this is?

6. To solve this problem, you need to give the client the proper credentials:
   a. On the laptop's **Config** tab, on the **Wireless0** page, enter the correct authentication information.
   b. If the connection does not re-establish, troubleshoot the problem.
   c. **Take a screenshot** of the information you entered on the wireless interface's configuration menu to establish successful authentication; submit this visual with your responses to this project's questions.

The wireless router offers many of the same GUI-based options you might see on a home router's user interface. To explore some of the security options, complete the following steps:

7. On the wireless router's **GUI** tab, what is the current IP address range? What is the client lease time?
8. On the **Wireless** > **Basic Wireless Settings** page, change the default SSID to **HappyVintage**. Scroll to the bottom of the screen and click **Save Settings**.
9. On the **Wireless** > **Wireless Security** page, WPA2 Personal is already selected. What other security mode options do you have? Which of these security modes is the most secure?
10. AES encryption is selected. What other encryption option do you have? Which of these encryption options is most secure?
11. Close the wireless router's configuration window. What changed on your network? Why do you think this is?
12. On the laptop's **Config** tab, on the **Wireless0** page, correct the SSID, and then close the laptop's configuration window. What happens on your network?
13. Make sure the wireless connection between the wireless router and the laptop is active. Save your project for use in Capstone Project 6-2 or continue with that project now.
14. Make some notes on your Wikidot website about your activities in Packet Tracer for this project.

Note to Instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site. Some Packet Tracer projects build on earlier Packet Tracer networks. If needed for one or more students, you can provide a previous project's solution file as a start file for one of these progression projects.

Capstone Project 6-2: Add IoT Devices to a Basic Wireless Network in Packet Tracer

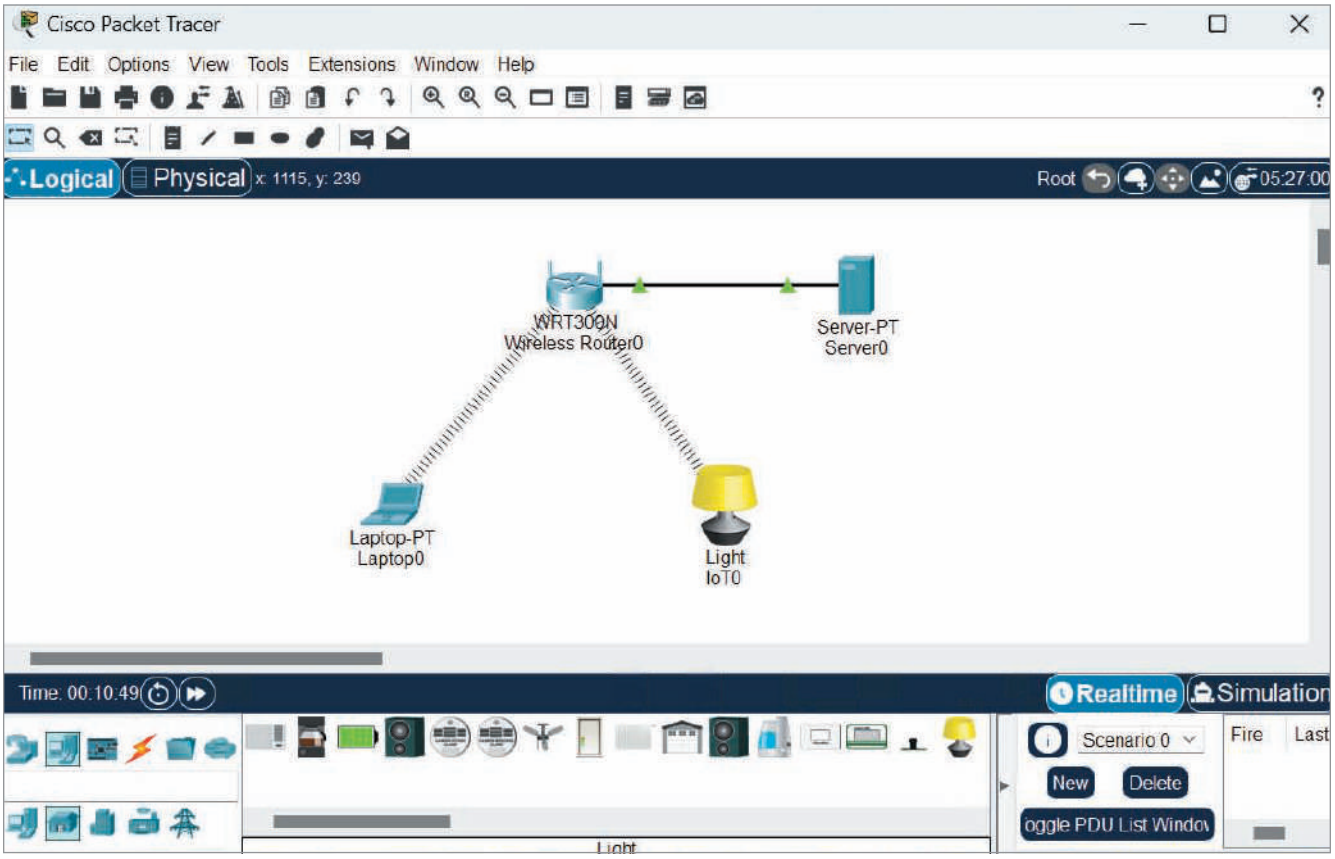
Estimated time: 45 minutes
Objective 4.1: Explain the importance of basic network security concepts.
Resources:
- Computer with Cisco Packet Tracer installed
- Packet Tracer network created in Capstone Project 6-1

Description:
This project picks up where Capstone Project 6-1 left off. In this project, you add some wireless IoT devices and configure them to be controlled remotely across your network. Beginning with the Packet Tracer network you created in Capstone Project 6-1, complete the following steps:

1. From the **End Devices** menu, insert a **Server**.
2. Using a **Copper Straight-Through** cable, connect Server0's **FastEthernet0** port to the wireless router's **Ethernet 1** port. Wait for the router-to-server connection to turn to green triangles on both ends.

3. On Server0's **Config** tab, on the **FastEthernet0** page, set the server to use DHCP to request an IP address assignment. What address is assigned to the server?
4. On Server0's **Services** tab, click **IoT** in the SERVICES pane and turn on the IoT network service. What other network services are listed in this pane that you recognize?
5. On Server0's **Desktop** tab, click **IoT Monitor**. Enter Server0's IP address that you noted in Step 3 and then click **Login**. You'll get an error message that you used the wrong username or password—this is because you have not yet created a user account. Click **Sign up now**. Enter **admin** for the username and password and then click **Create**. You are now logged into the IoT server.
6. In the Cisco Packet Tracer window, from the **End Devices** > **Home** menu, add the **Light** to your workspace.
7. On the light's **Config** tab on the **Wireless0** page, enter the correct information (SSID and authentication information) for the light to automatically connect wirelessly to your network (Figure 6-38). If the connection does not establish automatically, troubleshoot the problem. What authentication information did the light need to connect to the wireless router?

Figure 6-38 The IoT light connects wirelessly to the network



Source: Cisco Systems, Inc.

8. The light is connected to the network but is not yet communicating with the IoT server. To fix this, in the light's configuration window, click **Settings** in the left pane. Scroll down to the IoT Server section. Select **Remote Server**, and then enter the server's address, username, and password. Click **Connect**.

9. On the server's **Desktop** > **IoT Monitor** page, the light should now be listed under IoT Server - Devices. Position the server's configuration window so you can see the light in your workspace while changing the server's configuration. In the server's configuration window, click the **IoT0** entry to open the light's remote controls. Change the light's status. What happens to the light in your network?
10. Add a second IoT device of your choice. Make the needed changes to this device to connect it wirelessly to the network if possible or with a wired connection if necessary. Then register your new IoT device with the IoT server. Confirm that you can remotely control it from the server's IoT Monitor page.
11. Position the network devices and the IoT Monitor page so you can see all devices on the network and both IoT devices listed on the IoT Monitor page. **Take a screenshot** showing all this information; submit this visual with your responses to this project's questions.
12. You can save your project for future reference and experimentation; however, you will not need this network for future projects in this course.
13. Make some notes on your Wikidot website about your activities in Packet Tracer for this project.

Note to Instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site. Some Packet Tracer projects build on earlier Packet Tracer networks. If needed for one or more students, you can provide a previous project's solution file as a start file for one of these progression projects.

Solutions to Self-Check Questions

Section 6-1: Characteristics of Wireless Transmissions

1. Which of the following statements is false?
   Answer: a. Each Wi-Fi channel contains multiple frequency bands.
   Explanation: A band covers a frequency range that is further subdivided into channels; therefore, a **Wi-Fi channel cannot contain multiple frequency bands**. In the United States, the FCC has defined channels within the 2.4 GHz band, the 5 GHz band, and the 6 GHz band. Wi-Fi uses DSSS, which means its data streams are spread over all available frequencies within one of three, wide channels, all at the same time.

2. Which of the following statements is true?
   Answer: c. A Wi-Fi client connects to a range extender, which connects to an AP.
   Explanation: **A Wi-Fi client can connect to a wireless range extender, which can connect to an AP (access point)** to further extend the WLAN range. Satellite dishes are shaped to receive directional signals from one direction. Water in a fish aquarium and the glass around it will both refract waves as they travel into and through these different mediums. Cellular towers and cellular devices use omnidirectional antennas to improve mobility of cellular client devices, and a mobile device might remain connected to a single tower as the device moves, even if another tower becomes closer.

Section 6-2: 802.11 WLAN Standards

3. What was the first 802.11 standard to incorporate the 6 GHz band?
   Answer: a. Wi-Fi 6E
   Explanation: Beginning with **Wi-Fi 6E**, Wi-Fi networks can take advantage of the currently unlicensed 6 GHz frequency range. Older Wi-Fi standards are limited to the 2.4 and 5 GHz bands.

4. Which type of identifier allows wireless clients to roam freely from AP to AP?
Answer: c. ESSID
Explanation: Within an ESS (extended service set), a client can associate with any one of many APs that use the same **ESSID (ESS identifier)**. This allows users to roam, or change from AP to AP, without losing wireless network service. The identifier for a group of nodes using only one AP is a BSSID (basic service set identifier). The transmitter address field in an 802.11 header identifies the MAC address of the device transmitting a Wi-Fi message. IP addresses function at layer 3 and are not part of the Wi-Fi header.

Section 6-3: Implementing a Wi-Fi Network

5. Which device on the network requires the greatest throughput capacity?
Answer: c. Wireless bridge
Explanation: A **wireless bridge** connects two networks or two remote portions of one network; the throughput demands of a wireless bridge connected to another AP can be significantly higher than typical Wi-Fi clients. IoT devices such as a smart doorbell or security camera generally only need sufficient throughput to process commands, send updated sensor data, or possibly stream music or video—however, even streaming video will require less throughput than a bridge that supports multiple wireless clients. A wireless LAN controller might be used only initially to configure the APs, or the APs might remain connected to the wireless controller for continued management—however, management communication does not require much throughput capabilities.

6. Which offboarding policy can protect proprietary corporate information if a smartphone is lost?
Answer: a. Remote wipe
Explanation: The ability to perform a **remote wipe** is part of an offboarding policy that clears a device of important information, permissions, and apps without having physical access to the device. Onboarding gives the device trusted access to certain portions of the network. An onboarding agent might also scan the device for any needed OS or security updates, required security settings (such as timeouts or sign-in requirements), and any existing malware. A site survey assesses client requirements, facility characteristics, and coverage areas when determining the best placement for APs in a wireless network.

Section 6-4: Wi-Fi Network Security

7. Which Wi-Fi encryption standard was designed to use AES encryption?
Answer: b. WPA2
Explanation: CCMP, which is short for Counter Mode with CBC (Cipher Block Chaining) MAC (Message Authentication Code) Protocol, improves wireless security for newer devices that can use **WPA2 (Wi-Fi Protected Access, version 2)**. CCMP uses AES (Advanced Encryption Standard) to provide faster and more secure encryption than TKIP for wireless transmissions. WPA incorporates an encryption key generation and management scheme known as TKIP (Temporal Key Integrity Protocol) to improve security for legacy WEP-based devices. WEP (Wired Equivalent Privacy) offers two forms of authentication, neither of which is secure: OSA (Open System Authentication) and SKA (Shared Key Authentication). WEP2 does not exist.

8. Which standard adapted EAP to WLANs?
Answer: c. 802.1X
Explanation: EAP (Extensible Authentication Protocol) was originally designed to work only on point-to-point connections, usually on a WAN. However, it was adapted to work on both wired and wireless LANs in the **802.1X** standard, dubbed EAPoL (EAP over LAN). 802.11g is an early Wi-Fi standard. 802.11i was designed as a new wireless security standard to replace WEP and included the subset standard WPA. 802.3 is an Ethernet standard for wired networks.

Section 6-5: Troubleshooting Wi-Fi Networks

9. Which device will let you determine all frequencies within a scanned range that are present in a specific environment, not just Wi-Fi?
   Answer: d. Spectrum analyzer
   Explanation: A **spectrum analyzer** can assess the quality of a wireless signal by scanning a band of frequencies for signals and noise. A Wi-Fi analyzer can evaluate Wi-Fi network availability as well as help optimize Wi-Fi signal settings or help identify Wi-Fi security threats. A captive portal is the first page a new client displays in the browser when connecting to the guest network, and usually requires the user to agree to a set of terms and conditions before gaining further access to the guest network. A wireless LAN controller is used to configure and manage APs in a mesh network.

10. Which problem would a low RSSI level indicate?
    Answer: b. Insufficient wireless coverage
    Explanation: You can confirm whether **insufficient wireless coverage** is the problem by checking the RSSI (received signal strength indicator) level at the client's location. Mismatched standards will prevent a client from detecting an AP's broadcast. An incorrect passphrase or encryption protocol mismatch will prevent the completion of a client's authentication process.